# Glossary: every term, what it means, why it matters

47 terms in 8 groups · demo values: MARR 10%, inflation 5% (real MARR 4.76%), N = 25 years · 29 Oct 2026

How to read it: **Meaning** gives the plain definition and formula; **Course link** is where it is taught in CE 725; **In GreenLCC** says where the term appears in the tool and how it changes the decision.

## Time value of money

| Term | Meaning (and formula) | Course link | In GreenLCC / why it matters |
|---|---|---|---|
| **Time value of money (TVM)** | A rupee today is worth more than a rupee later, because of earning power, inflation and risk. | Ch 2 | The reason every future saving is discounted before it is compared with today's capex. |
| **MARR (discount rate)** | Minimum attractive rate of return: the hurdle rate the owner's money must earn. All cash flows are discounted at it. | Ch 5, Ch 7 | Set on Set up (demo 10% nominal). A higher MARR makes future savings worth less and favours cheap-upfront options. |
| **Nominal vs real rate (Fisher equation)** | (1 + i_nominal) = (1 + i_real)(1 + f). Real rate = (1 + MARR)/(1 + inflation) - 1. | Ch 2 | GreenLCC works in nominal rupees (prices escalate), so it discounts at the nominal MARR; the real MARR is shown for reference. |
| **Inflation (f)** | General rise in prices. Used to escalate O&M and the price of anything re-bought later. | Ch 2 | Demo 5%. Makes replacements later in life cost more in rupees of that year. |
| **Cash flow diagram (CFD)** | Timeline of money: up arrows = receipts/savings, down arrows = payments; t = 0 is today. | Ch 2 (Principle 4) | Compare measures -> 'Cash flow diagram' shows one measure year by year. |
| **Planning horizon / study period (N)** | One common analysis period for every alternative; cash flows after it are ignored and unused life is credited as salvage. | Ch 5 (method 2) | Demo N = 25 years. Lets options with unequal lives (LED 12 yrs, chiller 25 yrs) be compared fairly. |

## Interest factors

| Term | Meaning (and formula) | Course link | In GreenLCC / why it matters |
|---|---|---|---|
| **(F/P, i, n) and (P/F, i, n)** | Single payment: F = P(1+i)^n and P = F(1+i)^-n. | Ch 3 | (P/F) is the discount factor in the yearly table; (F/P, f, t) inflates a re-purchase price. |
| **(P/A, i, n) and (A/P, i, n)** | Uniform series: P = A[(1+i)^n - 1]/[i(1+i)^n]; (A/P) = 1/(P/A) is the capital recovery factor. | Ch 3, Ch 6 | (A/P, MARR, N) turns the life-cycle cost into EUAC. |
| **(A/F, i, n) sinking fund** | A = F · i/[(1+i)^n - 1]: the yearly amount that grows to F. | Ch 3, Ch 5 | Used in the capital-recovery cross-check CR = P(A/P) - S(A/F). |
| **Arithmetic gradient (P/G), (A/G)** | Series rising by a fixed amount G each year. | Ch 3 | Not needed by the model (prices rise by %), kept in core/finance.py for the textbook checks. |

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| **Geometric gradient (P/A1, g, i, n)** | Series A1, A1(1+g), A1(1+g)^2 …: P = A1[1 - ((1+g)/(1+i))^n]/(i - g). | Ch 3 (gradient series) | Electricity savings grow with tariff escalation and O&M grows with inflation; 'How it's calculated' hand-checks a measure with it. |

## Comparing alternatives

| Term | Meaning (and formula) | Course link | In GreenLCC / why it matters |
|---|---|---|---|
| **Present worth (PW) / life-cycle cost (LCC)** | PW = sum of (C_t - S_t)/(1+i)^t for t = 0…N: capex + O&M + replacements - salvage - savings, all at the MARR. | Ch 5 | The main number on every page. Choose the lowest PW of cost; negative means the measure pays for itself. |
| **Co-termination** | All alternatives must be compared over the same analysis period. | Ch 5 (requirement 1) | Guaranteed by the single study period N. |
| **Salvage / residual value** | Value left at the end of the horizon. GreenLCC uses last purchase price × remaining-life fraction. | Ch 5, Ch 6 | Credits a 30-year glazing for the 5 years it still has left at year 25. |
| **Replacement (re-purchase)** | When a measure's life (or its part's life) ends inside N it is bought again at today's price inflated to that year. | Ch 5 (LCM idea), Module 2 | LED (12 yrs) is bought at 0, 12 and 24; solar inverters every 12 years. |
| **Equivalent uniform annual cost (EUAC / EUAW)** | EUAC = PW × (A/P, i, N): the life-cycle cost as an equal yearly amount. | Ch 6 | Shown beside PW; it ranks options exactly like PW and is how budgets think (Rs per year). |
| **Capital recovery (CR)** | CR = (P - S)(A/P, i, n) + S·i: yearly cost of owning an asset bought for P and sold for S. | Ch 6 | Used as a unit-test cross-check of EUAC. |
| **Benefit-cost ratio (BCR)** | BCR = PW of savings ÷ PW of costs (capex + O&M + replacements - salvage). Accept if > 1. | Module 2 | Compare measures and the PDF shortlist. BCR > 1 exactly when PW of cost < 0. |
| **Discounted payback** | Years until the cumulative present worth turns in the owner's favour. | Module 2, Ch 1 (pitfall) | Replaces simple payback, which ignores the time value of money (the pitfall named in Ch 1). |
| **Incremental (do-nothing) baseline** | Each measure is costed as the difference from the code-minimum option it replaces. | Ch 7 (incremental analysis) | VRF capex = VRF price - split ACs it replaces. Makes every number an increment over doing nothing. |

## Rate of return

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| **Internal rate of return (IRR)** | The rate i* at which PW of the cash flow = 0. | Ch 7 | Shown per measure; accept if IRR >= MARR. Never used to rank (see ranking paradox). |
| **Incremental IRR (IROR)** | IRR of the difference between two alternatives (costlier - cheaper). If >= MARR, the extra money is worth spending. | Ch 7 (pairwise rule) | The headline on the Recommendation page: return on the extra capex of the recommended mix over the cheapest mix. |

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| **Simple vs non-simple cash flow; Descartes' rule** | A simple series changes sign once and has one IRR; each extra sign change can add another IRR root. | Ch 8 | Re-purchases (VRF at year 15, chiller overhauls) make many measures non-simple, so GreenLCC counts sign changes first. |
| **Modified IRR (MIRR)** | Outflows discounted at a finance rate, inflows compounded at a reinvestment rate; always one answer. | Ch 8 | Shown instead of IRR when the cash flow changes sign more than once (both rates = MARR). |
| **NPW vs IRR ranking paradox** | The option with the higher IRR can have the lower PW; IRR assumes reinvestment at the IRR itself. | Ch 8 | Why GreenLCC ranks on PW/EUAC and uses IRR/MIRR only as an accept-or-reject check. |

## Risk and uncertainty

| Term | Meaning (and formula) | Course link | In GreenLCC / why it matters |
|---|---|---|---|
| **Sensitivity analysis (tornado)** | Change one input at a time (±20%, ±2 pp) and see how much the life-cycle cost moves. | Module 3 | Risk page -> 'Sensitivity': tells you which inputs to verify first. |
| **Break-even point** | The value of an input at which two alternatives have equal PW, i.e. where the ranking flips. | Module 3 | Risk page: break-even MARR, tariff escalation and capex multiplier. Far from today's value = safe. |
| **MARR sweep** | PW of both routes plotted against the MARR; the crossing is the break-even MARR. | Module 3, Ch 7 | It sits close to the incremental rate of return: both answer 'how much return does the extra capex make?'. |
| **Monte Carlo simulation** | Draw thousands of futures (tariff growth, MARR, capex, solar yield) and recompute; count how often each route wins. | Module 3 | Risk page: win rate and the spread of life-cycle cost. Same seed = same numbers. |
| **P10 / P50 / P90** | 10th, 50th (median) and 90th percentile of the simulated results: 8 in 10 futures fall between P10 and P90. | Module 3 | Range quoted on the Risk page and in the PDF. |
| **Tariff escalation and its uncertainty (sd)** | Expected yearly growth of the grid tariff, and the spread used by the Monte Carlo. | Module 3 | Demo 5% ± 2 pp. Usually the input that moves the answer most. |

## Building and measure inputs

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| **Capex (incremental)** | Upfront price at year 0 above the code-minimum alternative. | Ch 5 | Measures sheet. The cheapest-upfront route minimises the sum of these. |
| **Life and replaceable part** | Service life of the measure; one major part with a shorter life (inverter, compressor, overhaul) and its cost. | Ch 5, Ch 6 | Drive re-purchases and salvage. Part replacement is skipped in a year the whole measure is re-bought. |
| **O&M (incremental)** | Extra yearly operation and maintenance cost vs the baseline, escalating at inflation. | Ch 5 | A chiller plant's operators and water treatment; negative O&M is not allowed (enter 0). |

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| **Energy and water saved** | kWh and kL a year saved vs the baseline, valued at the escalating tariffs. | - | Main benefit stream S_t; also drives the IGBC % thresholds. |
| **Degradation** | Yearly loss of savings or solar output (e.g. 0.5%/yr for PV modules). | - | Shrinks the savings series each year. |
| **Group (mutually exclusive alternatives)** | Options in one group are alternatives: at most one is chosen (one HVAC system, one solar size). | Ch 7 (mutually exclusive) | Defines the combinations the optimizer searches. |
| **Solar yield, self-consumption, export credit** | kWh per kWp a year; share used on site (worth the grid tariff); price paid for exported kWh. | - | Solar savings = self-used kWh × tariff + exported kWh × export credit (Rs 2.82, MERC FY25-26). |
| **Code-minimum baseline** | The building as it would be built anyway (ECBC-level). Every saving and cost is measured from it. | - | Baseline electricity and water use are entered on Set up. |

## IGBC and sustainability

| Term | Meaning (and formula) | Course link | In GreenLCC / why it matters |
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| **IGBC credit, points, level** | Indian Green Building Council rating: credits earn points; Certified/Silver/Gold/Platinum need 50/60/70/80 (illustrative). | Module 5 (green finance) | The target level is the constraint the optimizer must meet. |
| **Fixed vs threshold credits; mandatory credit** | Fixed credits add set points; threshold credits score the combined % saving; a mandatory credit must earn > 0 points. | - | Combined scoring means savings are never double counted. |
| **Intangibles alongside PW** | Benefits PW cannot price (rating, comfort, emissions) are weighed next to the money result. | Ch 5 (requirement 4) | IGBC points and $CO_2$ are shown beside PW, not converted into rupees. |
| **$CO_2$ avoided / grid emission factor** | Lifetime kWh saved × $tCO_2$ per MWh (CEA database, 0.727 for FY 2023-24). | Module 5 | One KPI per mix; relevant to green finance and ESG reporting. |

## Tool outputs

| Term | Meaning (and formula) | Course link | In GreenLCC / why it matters |
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| **Cheapest-upfront route** | The mix that reaches the target for the least capex: today's L1 habit. | Ch 1 (choice) | The 'defender' the recommendation is compared with. |
| **Recommended route** | The mix that reaches the target at the lowest life-cycle cost; found by checking every combination. | Ch 5, Ch 7 | The 'challenger'. It wins when its extra capex earns at least the MARR. |
| **Upgrade path** | Lowest-LCC mix at each IGBC level, and the return on stepping up from the level below. | Ch 7 (pairwise elimination) | Shows when a higher rating is free over the building's life. |
| **Decision report, results workbook, input workbook** | 6-page PDF for the owner; Excel with every number unrounded; Excel inputs to edit and reopen. | - | Report page downloads. |